
Chapter 2 ~ SPATIAL AND INTENSITY RESOLUTION

Created By

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Image Sampling and Quantization

- ✓ **Image sampling** involves taking the value of the image at regular spatial intervals. The length of the intervals defines the spatial resolution of the image.
- ✓ If we increase the number of pixels in the image, we can better represent an analog object into a digital image.
- ✓ **Image quantization** involves discretizing the intensity values of the analog image.
- ✓ If we increase the number of discrete values representing the image intensity values in the image, the better we can represent an analog object into a digital image.

Image Sampling and Quantization

- ✓ **Widely used definition of image resolution** is the largest number of discernible **line pairs per unit distance**.
- ✓ In the printing and publishing industry, **dots per unit distance** is a measure of image resolution used.
 - ✓ Example: **In the U.S.**, this measure usually is expressed as *dots per inch(dpi)*.
- ✓ **Sampling** is the principle factor determining the **spatial resolution of an image**.

Spatial Resolution

- ❖ **Intuitively, spatial resolution** is a measure of the smallest discernible detail in an image.
- ❖ **Quantitatively, spatial resolution** can be stated in a number of ways, with line pairs per unit distance, and dots (pixels) per unit distance being among the most common measures.

Intensity Resolution

- **Intensity (Gray-Level) resolution** similarly refers to the smallest **discernible change** in intensity level.
- It is common practice to refer to **the number of bits** used to quantize intensity as the intensity resolution
- For example, it is common to say that an image whose intensity is quantized into 256 levels has **8 bits of intensity resolution**.

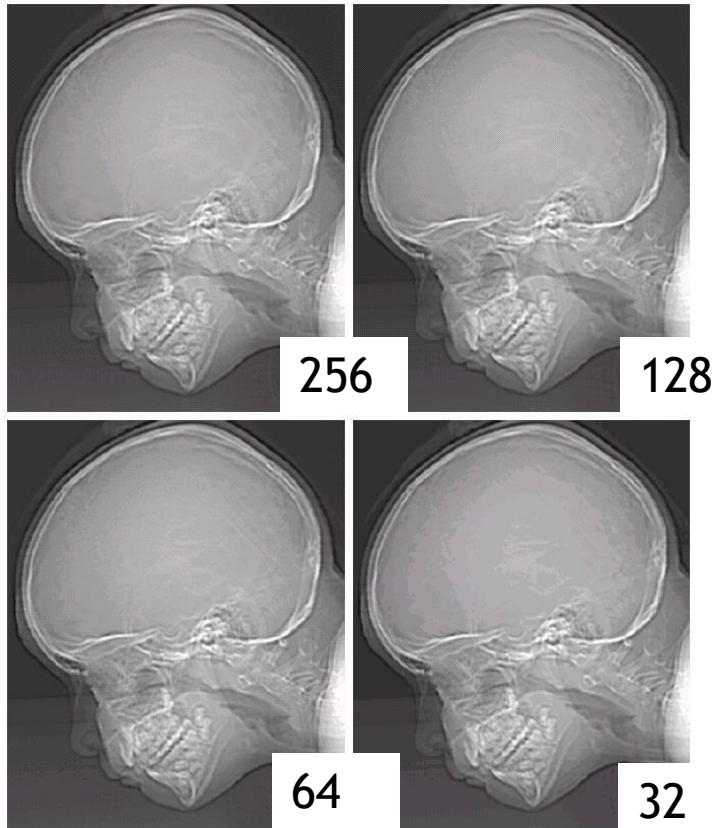
Illustration of the effects of reducing image spatial resolution

- ❖ Figure shows the effects of reducing spatial resolution in an image.
- ❖ The images in Figs. (a) through (d) are shown in **1250, 300, 150, and 72 dpi**, respectively.
- ❖ **Naturally**, the lower resolution images are smaller than the original.
- ❖ For example, the original image is of size 3692 X 2812 pixels, but the 72 dpi image is an array of size 213 X 162 pixels.

✓ It is possible to improve on the results of Fig. (d) by the choice of **interpolation method** used.

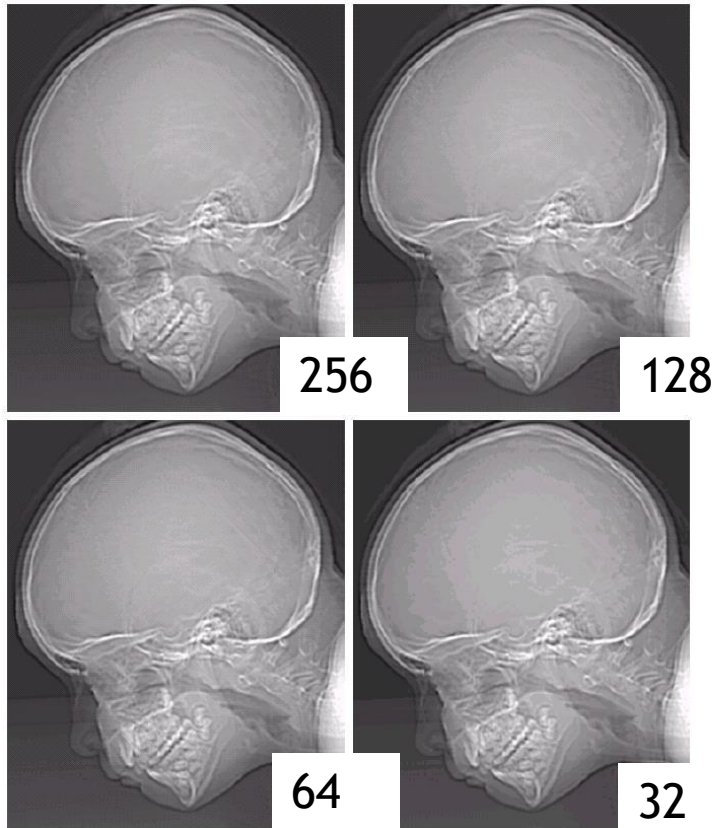


Typical effects of varying the number of intensity levels in a digital image



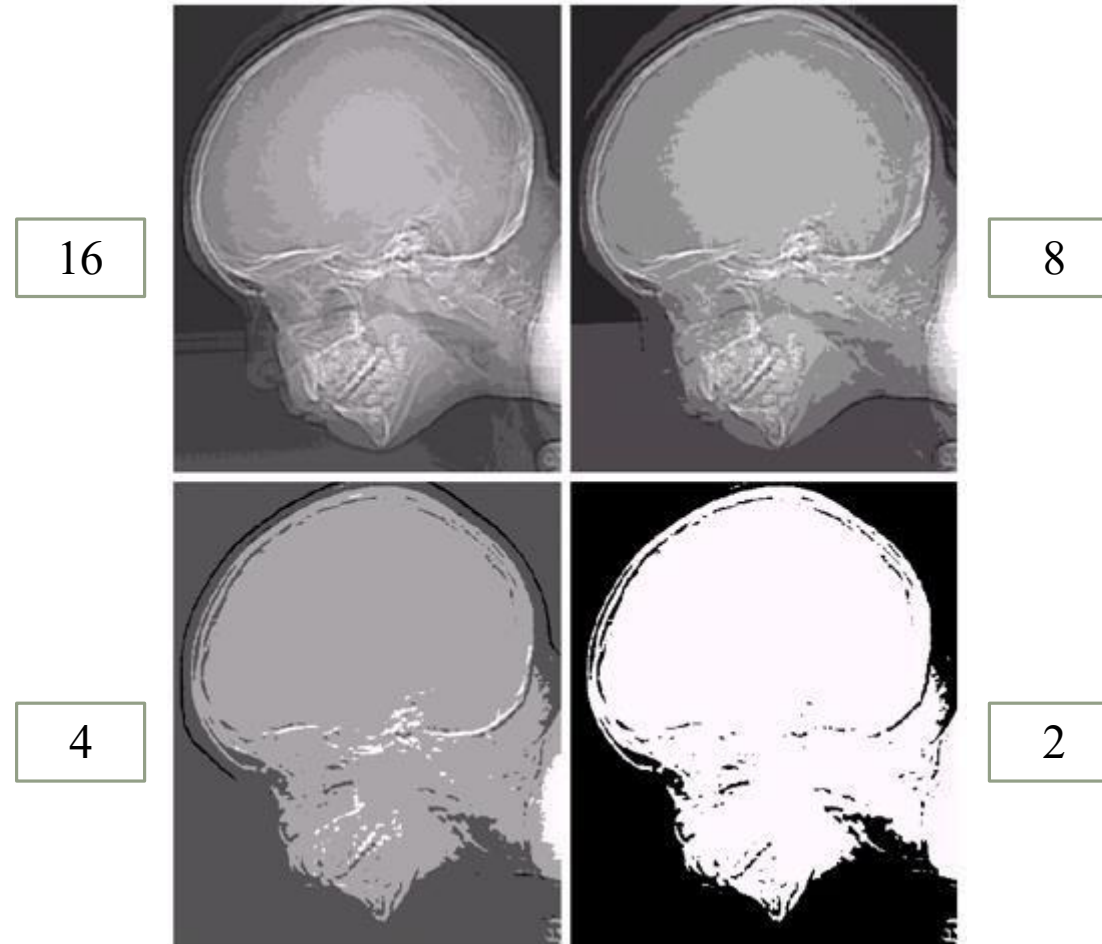
- ❖ In this example, we keep the number of samples constant and **reduce the number of intensity levels from 256 to 32**, in integer powers of 2.
- ❖ Here, all the images are in size of 452 X 374 pixels.

Typical effects of varying the number of intensity levels in a digital image



❖ The 32-level image, however, has an **imperceptible** set of very **fine ridge-like structures** in areas of constant or nearly constant intensity (**particularly in the skull**). This effect, caused by the use of an **insufficient number of intensity levels** in smooth areas of a digital image, is **called false contouring**.

Typical effects of varying the number of intensity levels in a digital image



Spatial and Intensity Resolution

- ✓ **Huang [1965]** attempted to quantify experimentally the effects on image quality produced by **varying N and k** simultaneously.

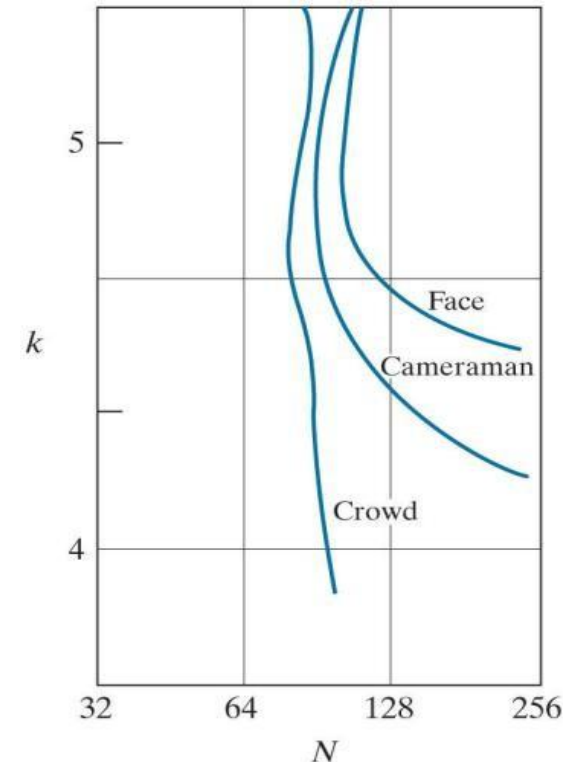
Dataset



Figure 1: (a) Image with a low level of detail. (b) Image with a medium level of detail. (c) Image with a relatively large amount of detail

Spatial and Intensity Resolution

- ✓ Sets of these three types of images were generated by varying N and k , and **observers** were then asked to rank them according to their subjective quality.
- ✓ Results were summarized in the form of so-called *iso-preference curves* in the NK -plane.
- ✓ The **key point** of interest in the context of the present discussion is that **iso-preference curves** tend to become more vertical as the detail in the image increases.



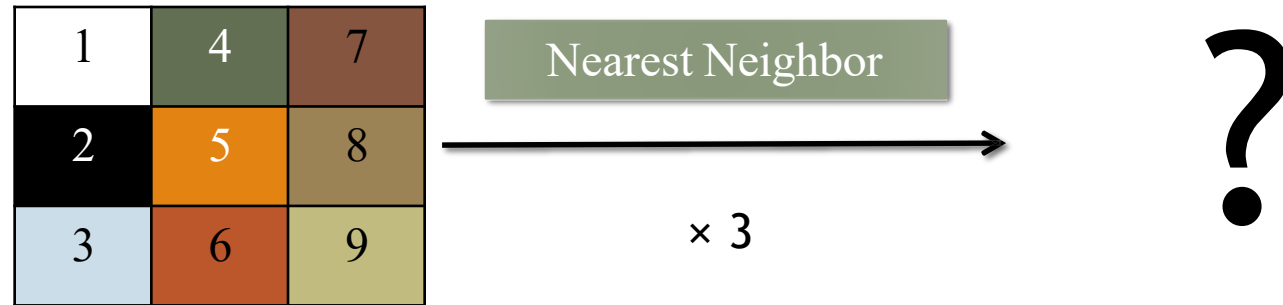
Images with a large amount of detail only a few intensity levels may be needed.

Larger values for N and k , which implies better picture quality.....

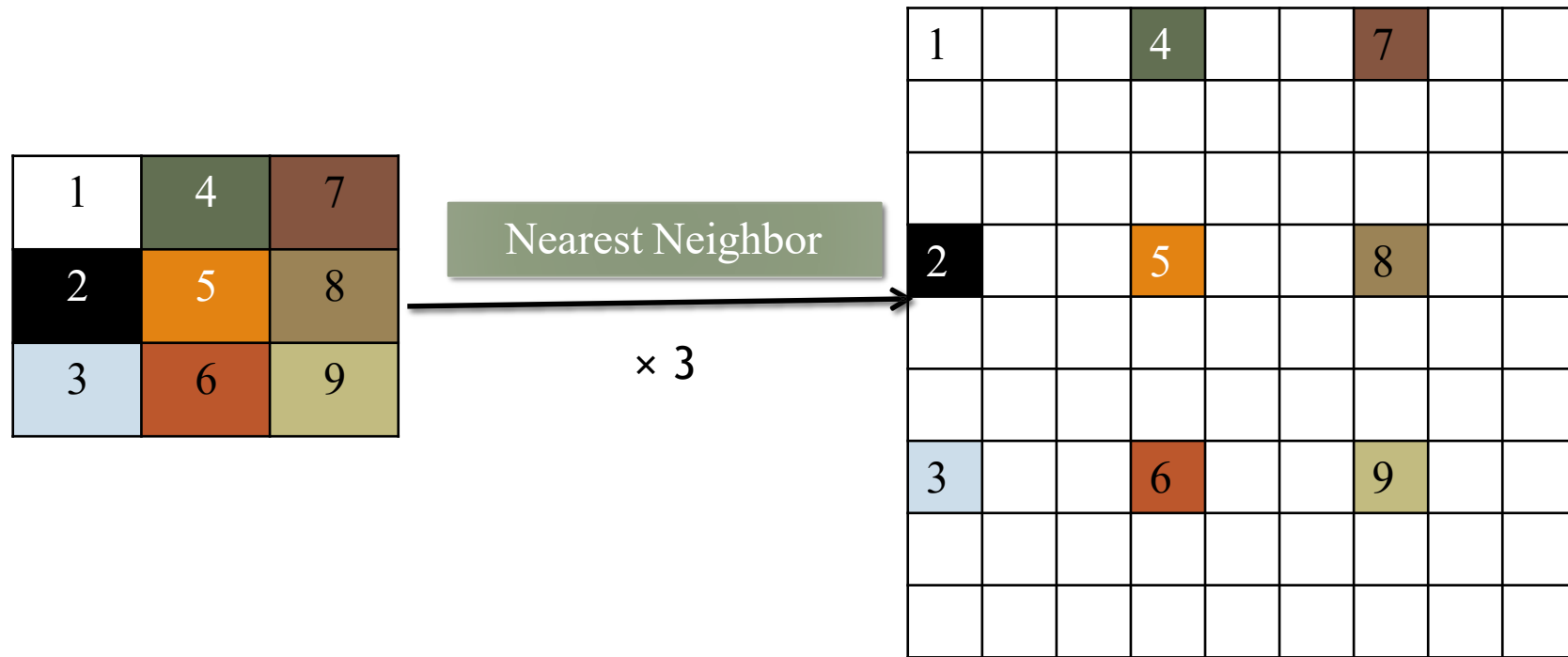
Image Interpolation

- ✓ **Zooming, shrinking**, rotating, and geometric corrections.
- ✓ **Interpolation** is the process of using known data to estimate values at unknown locations.
 - ❖ **Nearest Neighbor Interpolation**
 - ❖ **Bilinear Interpolation**
 - ❖ **Bicubic Interpolation**
 - ✓ Bicubic interpolation is the standard used in commercial image editing programs, such as Adobe Photoshop and Corel Photo paint.

Nearest Neighbor Interpolation



Nearest Neighbor Interpolation



Nearest Neighbor Interpolation

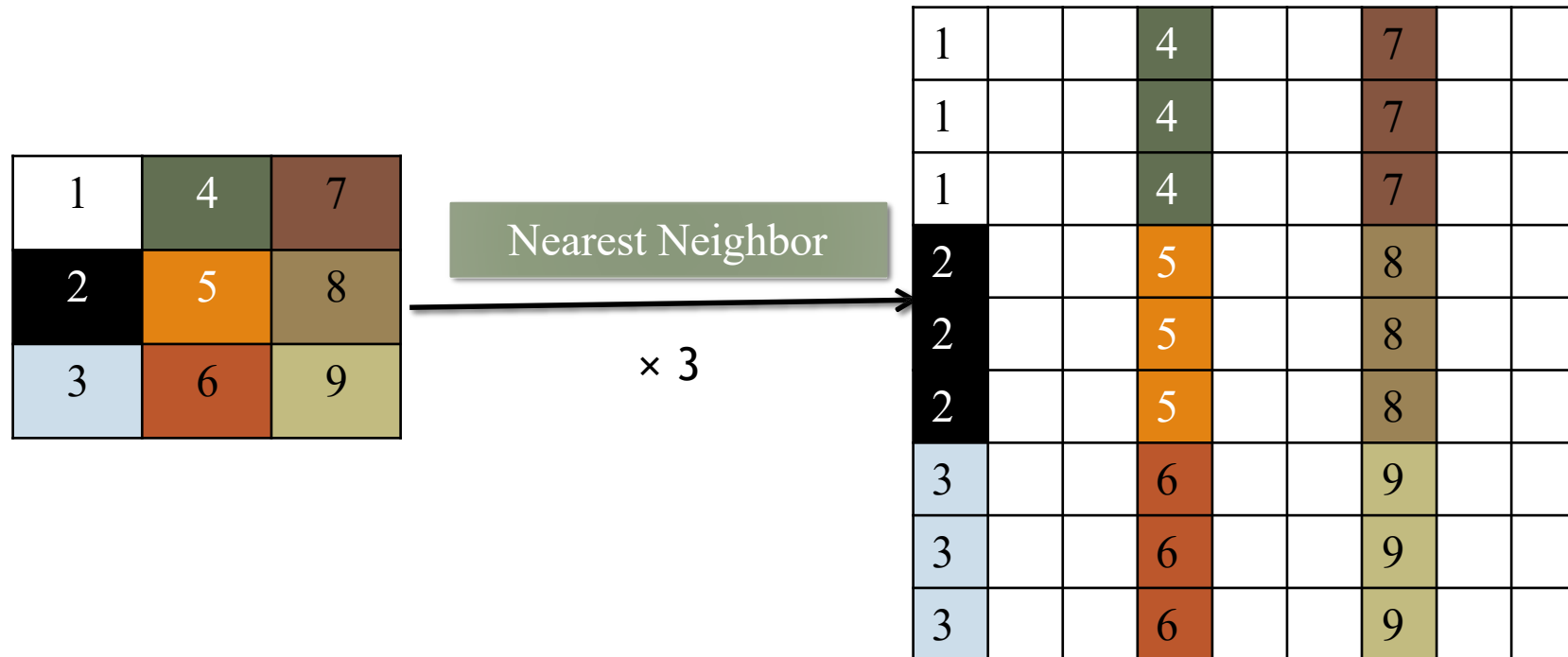
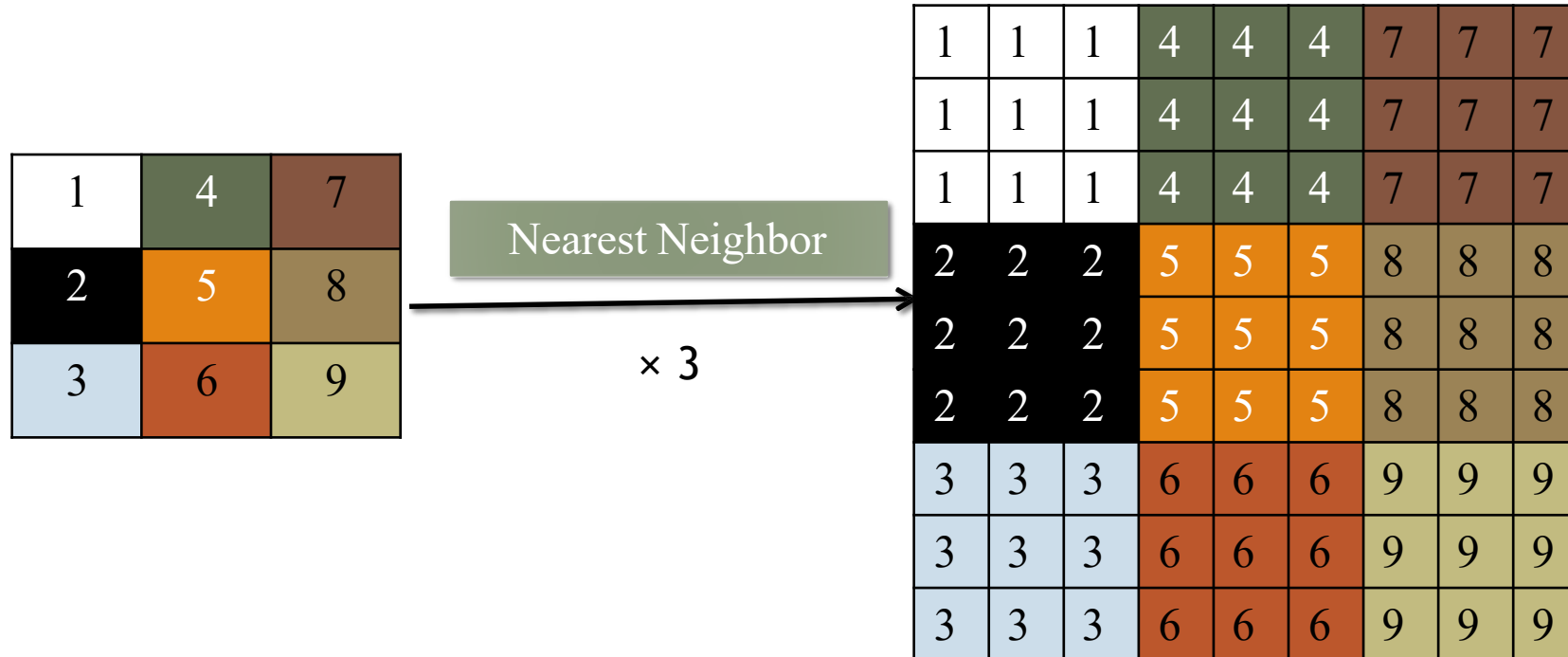
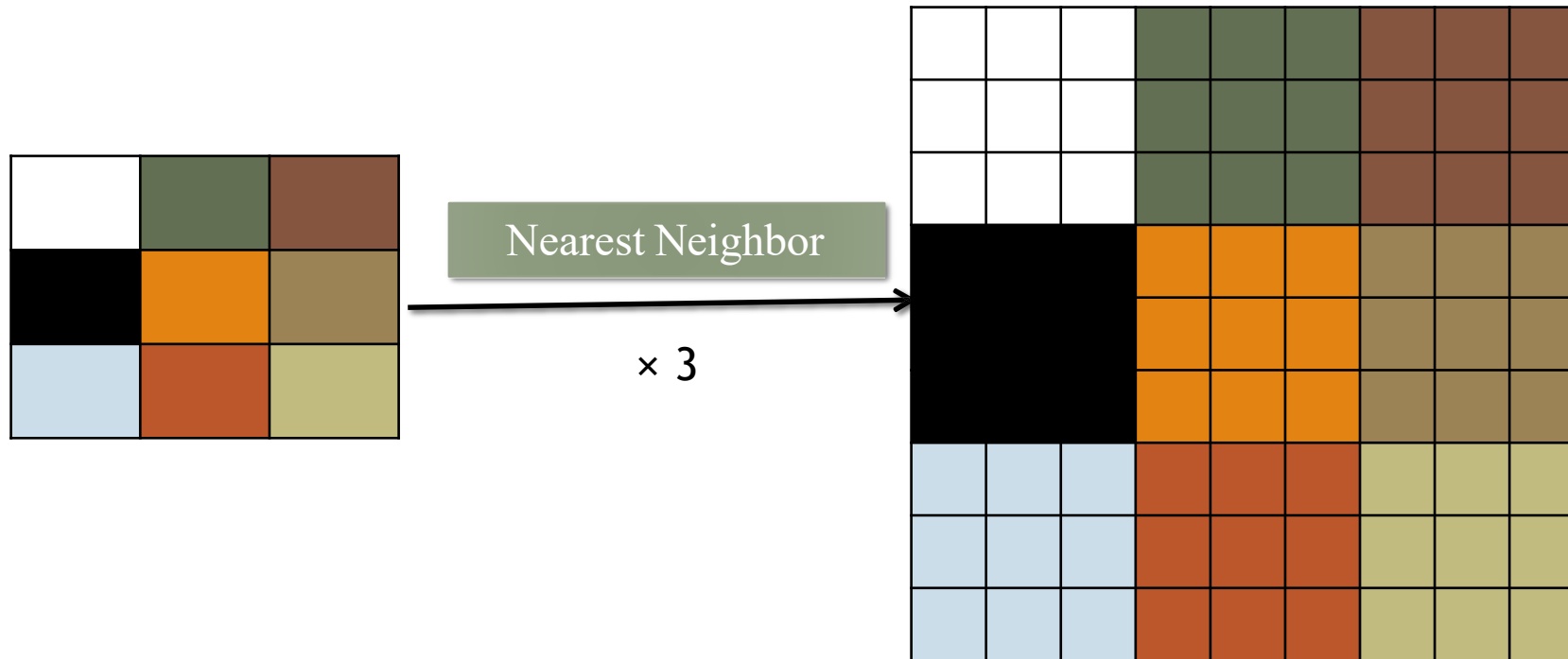


Image Sampling and Quantization

Nearest Neighbor Interpolation



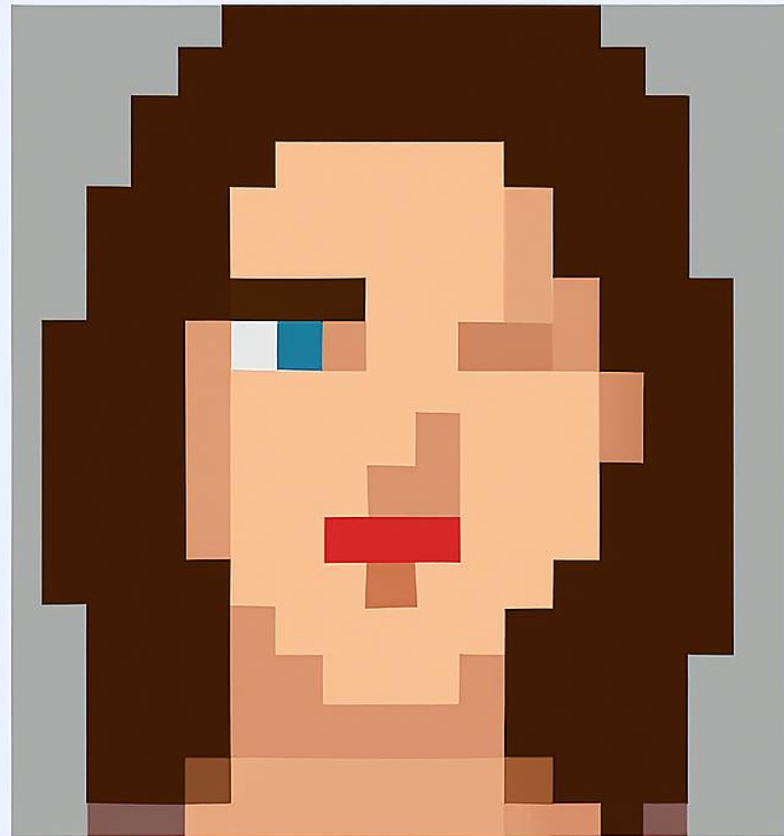
Nearest Neighbor Interpolation



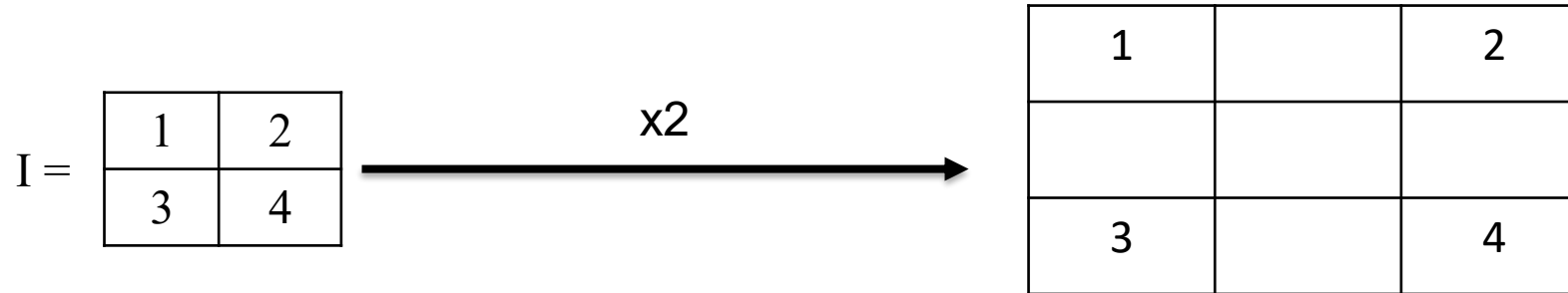
Original Image



**Interpolated with
Nearest Neighbors**



Bilinear Interpolation



$$\begin{aligned}\text{Size of scaled image} &= (n - 1) * k + 1 \times (m - 1) * k + 1 \\ &= (2 - 1) * 2 + 1 \times (2 - 1) * 2 + 1 \\ &= 3 * 3\end{aligned}$$

Step 1: Horizontal interpolation (2 intervals)

- Between 1 and 2: $1 \times 1/2 + 2 \times 1/2 = 1.5$
- Between 3 and 4: $3 \times 1/2 + 4 \times 1/2 = 3.5$

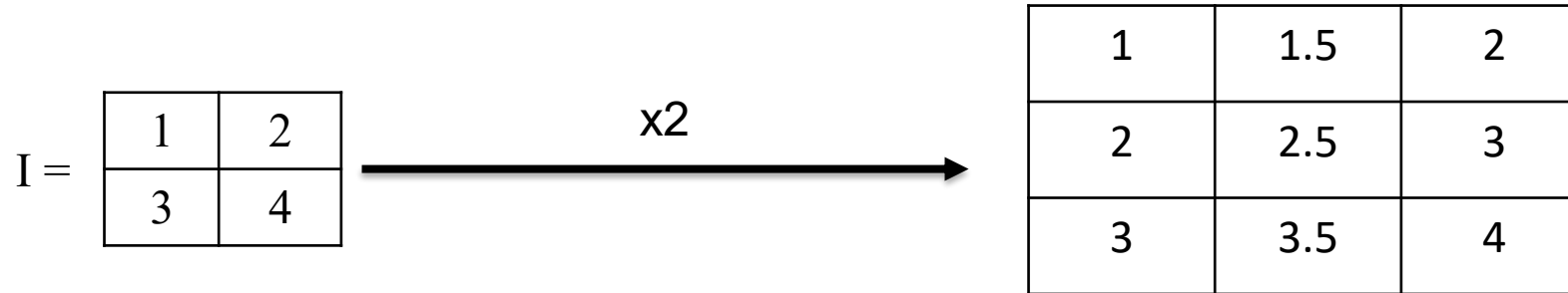
Result after horizontal fill:

$\begin{bmatrix} 1 & 1.5 & 2 \\ 3 & 3.5 & 4 \end{bmatrix}$

Step 2: Vertical interpolation (2 intervals)

- Between 1 and 3: $1 \times 1/2 + 3 \times 1/2 = 2$
- Between 1.5 and 3.5: $1.5 \times 1/2 + 3.5 \times 1/2 = 2.5$
- Between 2 and 4: $2 \times 1/2 + 4 \times 1/2 = 3$

Bilinear Interpolation



Step 1: Horizontal interpolation (2 intervals)

- Between 1 and 2: $1 \times 1/2 + 2 \times 1/2 = 1.5$
- Between 3 and 4: $3 \times 1/2 + 4 \times 1/2 = 3.5$

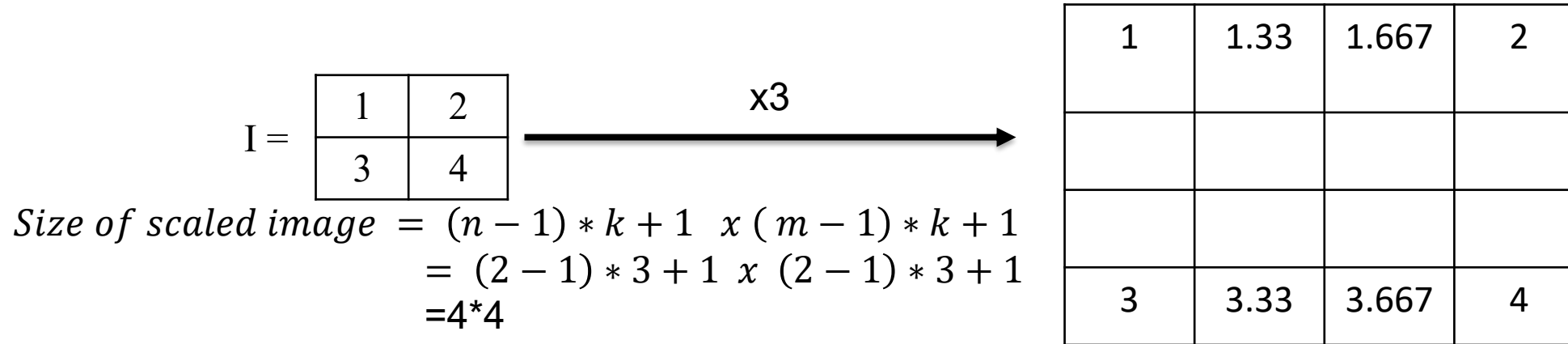
Result after horizontal fill:

$\begin{bmatrix} 1 & 1.5 & 2 \\ 3 & 3.5 & 4 \end{bmatrix}$

Step 2: Vertical interpolation (2 intervals)

- Between 1 and 3: $1 \times 1/2 + 3 \times 1/2 = 2$
- Between 1.5 and 3.5: $1.5 \times 1/2 + 3.5 \times 1/2 = 2.5$
- Between 2 and 4: $2 \times 1/2 + 4 \times 1/2 = 3$

Bilinear Interpolation



Step 1: Horizontal interpolation (3 intervals)

Between 1 and 2:

$$1 \times 2/3 + 2 \times 1/3 = 1.33$$

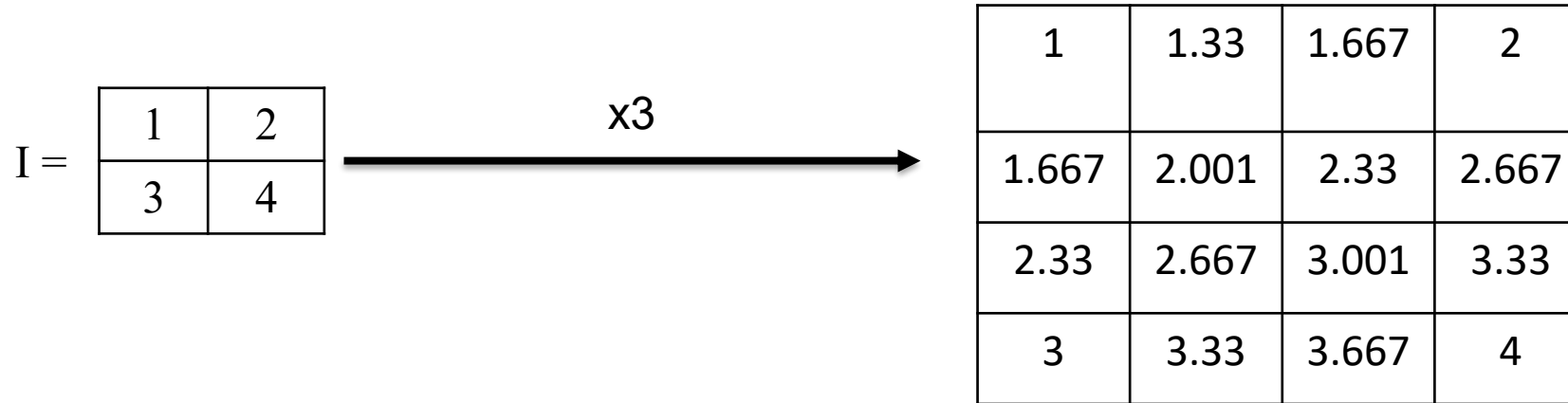
$$1 \times 1/3 + 2 \times 2/3 = 1.667$$

Between 3 and 4:

$$3 \times 2/3 + 4 \times 1/3 = 3.33$$

$$3 \times 1/3 + 4 \times 2/3 = 3.667$$

Bilinear Interpolation



Step 2: Vertical interpolation

1. Column 1 (1 and 3):

- New row 1: $1 \times 2/3 + 3 \times 1/3 = 1.667$
- New row 2: $1 \times 1/3 + 3 \times 2/3 = 2.333$

2. Column 2 (1.33 and 3.33):

- New row 1: $1.33 \times 2/3 + 3.33 \times 1/3 = 2$
- New row 2: $1.33 \times 1/3 + 3.33 \times 2/3 = 2.667$

3. Column 3 (1.667 and 3.667):

- New row 1: $1.667 \times 2/3 + 3.667 \times 1/3 = 2.334$
- New row 2: $1.667 \times 1/3 + 3.667 \times 2/3 = 3.001$

4. Column 4 (2 and 4):

- New row 1: $2 \times 2/3 + 4 \times 1/3 = 2.667$
- New row 2: $2 \times 1/3 + 4 \times 2/3 = 3.333$

Bilinear Interpolation

I =

1	2
3	4

```
>> imresize(I,2,'bilinear')
```

I =

1.00	1.25	1.75	2.00
1.50	1.75	2.25	2.50
2.50	2.75	3.25	3.50
3.00	3.25	3.75	4.00

$$\frac{3}{4} \times 1 + \frac{1}{4} \times 2 = \frac{5}{4} = 1.25$$

I =

1.00			2.00
3.00			4.00

Bilinear Interpolation

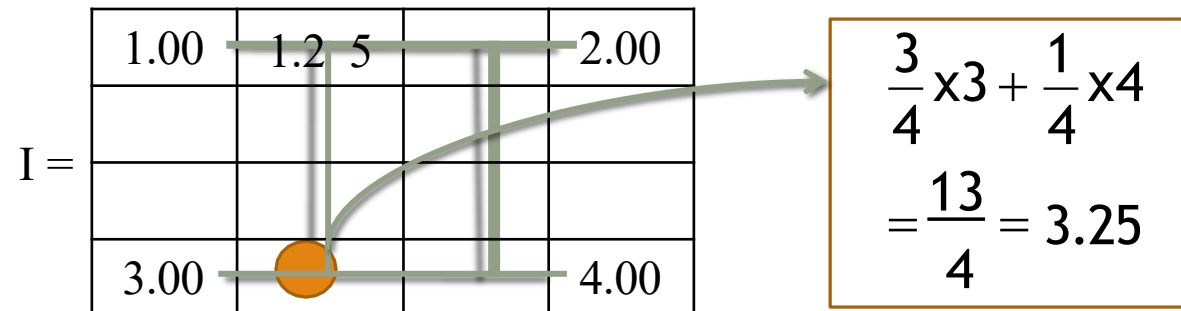
I =

1	2
3	4

```
>> imresize(I,2,'bilinear')
```

I =

1.00	1.25	1.75	2.00
1.50	1.75	2.25	2.50
2.50	2.75	3.25	3.50
3.00	3.25	3.75	4.00



Bilinear Interpolation

I =

1	2
3	4

```
>> imresize(I,2,'bilinear')
```

I =

1.00	1.25	1.75	2.00
1.50	1.75	2.25	2.50
2.50	2.75	3.25	3.50
3.00	3.25	3.75	4.00

I =

1.00	1.25		2.00
3.00	3.25		4.00

$$\frac{3}{4} \times 1.25 + \frac{1}{4} \times 3.25$$
$$= \frac{7}{4} = 1.75$$

Bilinear Interpolation

I =

1	2
3	4

```
>> imresize(I,2,'bilinear')
```

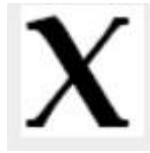
I =

1.00	1.25	1.75	2.00
1.50	1.75	2.25	2.50
2.50	2.75	3.25	3.50
3.00	3.25	3.75	4.00

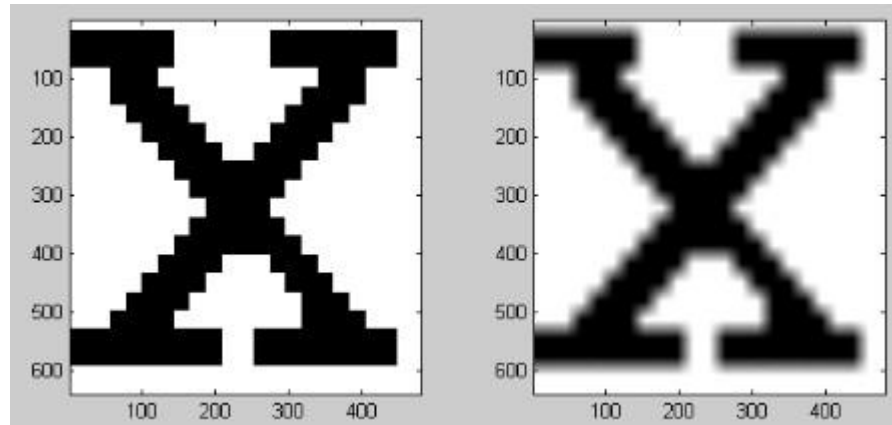
I =

1.00	1.25	?	2.00
?	1.75	?	?
?	?	?	?
3.00	3.25	?	4.00

Zooming in Interpolation



Original image



Nearest Neighbor

Bi-Linear

Bilinear Interpolation

1	4	7
2	5	8
3	6	9

x3



1			4			7
2			5			8
3			6			9

$$\begin{aligned}\text{Size of scaled image} &= (n - 1) * k + 1 \times (m - 1) * k + 1 \\ &= (3 - 1) * 3 + 1 \times (3 - 1) * 3 + 1 \\ &= 7 \times 7\end{aligned}$$

Bilinear Interpolation

1	4	7
2	5	8
3	6	9

x3



1	2	3	4	5	6	7
2			5			8
3			6			9

**Horizontal interpolation
(3 intervals)**

Between 1 and 4:

$$1 \times 2/3 + 4 \times 1/3 = 2$$

$$1 \times 1/3 + 4 \times 2/3 = 3$$

Between 4 and 7:

$$4 \times 2/3 + 7 \times 1/3 = 5$$

$$4 \times 1/3 + 7 \times 2/3 = 6$$

Bilinear Interpolation

1	4	7
2	5	8
3	6	9

x3



1	2	3	4	5	6	7
2	3	4	5	6	7	8
3			6			9

**Horizontal interpolation
(3 intervals)**

Between 1 and 4:

$$1 \times 2/3 + 4 \times 1/3 = 2$$

$$1 \times 1/3 + 4 \times 2/3 = 3$$

Between 4 and 7:

$$4 \times 2/3 + 7 \times 1/3 = 5$$

$$4 \times 1/3 + 7 \times 2/3 = 6$$

**Horizontal interpolation
(3 intervals)**

Between 2 and 5:

$$2 \times 2/3 + 5 \times 1/3 = 3$$

$$2 \times 1/3 + 5 \times 2/3 = 4$$

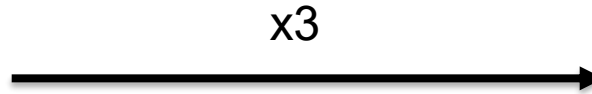
Between 5 and 8:

$$5 \times 2/3 + 8 \times 1/3 = 6$$

$$5 \times 1/3 + 8 \times 2/3 = 7$$

Bilinear Interpolation

1	4	7
2	5	8
3	6	9



1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9

**Horizontal interpolation
(3 intervals)**

Between 1 and 4:

$$1 \times 2/3 + 4 \times 1/3 = 2$$

$$1 \times 1/3 + 4 \times 2/3 = 3$$

Between 4 and 7:

$$4 \times 2/3 + 7 \times 1/3 = 5$$

$$4 \times 1/3 + 7 \times 2/3 = 6$$

**Horizontal interpolation
(3 intervals)**

Between 2 and 5:

$$2 \times 2/3 + 5 \times 1/3 = 3$$

$$2 \times 1/3 + 5 \times 2/3 = 4$$

Between 5 and 8:

$$5 \times 2/3 + 8 \times 1/3 = 6$$

$$5 \times 1/3 + 8 \times 2/3 = 7$$

**Horizontal interpolation
(3 intervals)**

Between 3 and 6:

$$3 \times 2/3 + 6 \times 1/3 = 4$$

$$3 \times 1/3 + 6 \times 2/3 = 5$$

Between 6 and 9:

$$6 \times 2/3 + 9 \times 1/3 = 7$$

$$6 \times 1/3 + 9 \times 2/3 = 8$$

Bilinear Interpolation

1	4	7
2	5	8
3	6	9

x3



1	2	3	4	5	6	7
1.33						
1.67						
2	3	4	5	6	7	8
2.33						
2.667						
3	4	5	6	7	8	9

**Vertical interpolation
(3 intervals)**

Between 1 and 2:

$$1 \times 2/3 + 2 \times 1/3 = 2$$

$$1 \times 1/3 + 2 \times 2/3 = 3$$

Between 2 and 3:

$$2 \times 2/3 + 3 \times 1/3 = 5$$

$$2 \times 1/3 + 3 \times 2/3 = 6$$

Bilinear Interpolation

1	4	7
2	5	8
3	6	9

x3



1	2	3	4	5	6	7
1.33	2.33	3.33	4.33	5.33	6.33	7.33
1.67	2.67	3.67	4.67	5.67	6.67	7.67
2	3	4	5	6	7	8
2.33	3.33	4.33	5.33	6.33	7.33	8.33
2.667	3.67	4.67	5.67	6.67	7.67	8.67
3	4	5	6	7	8	9

Vertical interpolation
Similarly for other
points.

Thank You ☺